AMERY ICE SHELF BOREHOLE MICROCAT AND ADCP DATA, 2011/2012 SEASON - DATA PROCESSING AND QUALITY

Mark Rosenberg (data processor)
Amery Ice Shelf borehole drill team (data collectors)

INTRODUCTION

This report summarises the latest AMISOR oceanographic data, season 2011/2012, collected from oceanographic equipment deployed through the Amery Ice Shelf boreholes. Time series data of temperature, salinity and pressure were downloaded from three strings of Seabird SBE37IM microcats at AMISOR boreholes AM03 (6 months of data), AM05 (12.5 months of data) and AM06 (12.5 months of data). Data were not downloaded from borehole AM04. Three current profiles were obtained from an ADCP (RDI 300 kHz broadband Workhorse) on the AM06 string. Data processing and data quality details are included in the report.

Table 1: Station details for AMISOR season 2011/2012 borehole microcats and ADCP. Note that borehole absolute locations change in time, as the ice shelf is in motion. For the microcats, positive clock errors indicate microcat clocks were fast. Instrument pressures/depths are mean values from the pressure sensors over the data dates, including all pressure steps. For microcat 7325, this mean is after correction of pressure data. For microcat 7327, the pressure/depth is a point value from soon after initial deployment. Note that data were not downloaded from microcat 3883 at AM03 (flat batteries); and no data were downloaded from AM04 microcats. Mean ADCP depth/pressure is from ADCP pressure sensor data (calibration may be out).

Borehole microcats

borehole	position at given date		microcat	instrument pos.		data dates		rec.	clock
	latitude	longitude		depth (m)	pressure (dbar)	start	end	interval (min.)	error (sec.)
AM03	70°32.468'S	70°20.625'E	4055	651.5	659.2	09-Jan-2011	11-Jul-2011	30	+172
AM03	(on 18-Jan-2012)		4054	859.3	869.9	09-Jan-2011	03-Jul-2011	30	+132
AM03	"	"	3883	-	-	-	-	-	-
AM05	70°13.539'S	69°41.137'E	7327	573.5	580.2	05-Feb-2011	24-Feb-2012	30	+6
AM05	(on 24-Feb-2012)		7326	718.8	727.4	05-Feb-2011	24-Feb-2012	30	+25
AM05	"	"	7322	854.3	864.9	05-Feb-2011	24-Feb-2012	30	+13
AM06	70°14.373'S	71°22.379'E	7325	545.9	552.2	05-Feb-2011	24-Feb-2012	30	+20
AM06	(on 24-Feb-2012)		7324	672.1	680.1	05-Feb-2011	24-Feb-2012	30	+29
AM06	"	"	7323	780.7	790.2	05-Feb-2011	24-Feb-2012	30	+14
AM06	"	"	ADCP	618.6	626.1	23-Jan-2012	23-Jan-2012	30	-

Table 2: Microcat pressure sensor ratings

borehole	microcat	sensor rating (dbar)	borehole	microcat	sensor rating (dbar)
AM01	1969	1000	AM04	1972	1000
AM01	1970	1000	AM04	1973	1000
AM01	1971	1000	AM04	1974	3500
AM02	1623	1034	AM05	7327	1000
AM02	1624	1034	AM05	7326	1000
AM02	1174	2068	AM05	7322	2000
AM03	4055	1000	AM06	7325	1000
AM03	4054	1000	AM06	7324	1000
AM03	3883	2000	AM06	7323	2000

MICROCAT DATA

The microcats were downloaded by Adam Christensen (no field report available). The microcat moorings have already been described in previous field and data reports. This was the second download of data from the AM05 and AM06 instruments (deployed in December 2009 and January 2010 respectively), and the sixth download from AM03 instruments (deployed in December 2005). All latest microcat data downloaded during the 2011/2012 season (Table 1) were assigned the same decimal time scheme consistent with previous data i.e. decimal days as counted from midnight on December 31st 2000 (so, e.g. midday on January 1st 2001 = 0.5 decimal time; midday on January 1st 2002 = 365.5).

The microcats were downloaded using the SeaBird terminal program Seaterm, and instrument clock errors (Table 1) were obtained from the "debug" instrument communication files. Clock drift corrections were applied to the AM03 data only (clock errors for AM05 and AM06 were small). For AM05 and AM06 microcats, microcat clocks were all reset to GPS time following data download and prior to reinitialisation of logging. For AM03, there are no field notes or instrument communication files to confirm whether clocks were reset. Clock check and clock set information for the ADCP is unknown.

Manufacturer supplied calibrations (June 2005 for AM03 instruments and September 2009 for AM05 and AM06) were applied internally by the microcats, and calibrated data were output. The data were then processed, as follows:

* The files were padded at the start and end, data gaps were checked for and filled, and decimal times were calculated.

* For AM03 instruments, decimal times were "compressed" linearly throughout the time series to correct for clock error (Table 1).

* Mean pressure/depth values were calculated for the year's data for each microcat (Table 1), with all pressure steps included in the means.

* Pressure data corrections were required for the shallowest microcat at AM05 and AM06 (#7327 and #7325 respectively), and conductivity/salinity data were recalculated with the corrected pressures (more details given below).

Microcat data quality notes

A whole range of data features are present in the current season's download, from interesting real data features, to suspect but unexplained data, to obviously bad data.

The following data spikes are present in the data (not removed from the files):

borehole	microcat	spike	times
AM03	4055	cond/sal fouling spike	1100 to 1900 on 07/02/2011
AM03	4054	cond/sal fouling spike	1630 to 1800 on 10/01/2011
AM03	4054	cond/sal fouling spike	1600 to 1800 on 24/01/2011
AM03	4054	cond/sal fouling spike	1330 on 13/03/2011
AM03	4054	temp/sal spike	1930 on 25/05/2011
AM05	7327	cond/sal fouling spike	1400 on 28/03/2011 to 0530 on 29/03/2011
AM05	7322	cond/sal fouling spike	1630 on 01/09/2011
AM05	7322	cond/sal fouling spike	2030 to 2200 on 25/10/2011
AM05	7322	cond/sal fouling spike	0600 on 25/11/2011

Microcat data are plotted in Figures 1 to 5. From the pressure sensor data (Figure 4), iceshelf deformation events are evident at AM05, as indicated by pressure steps. Strong current events, causing temporary "push-up" of the instruments, are largest at AM05 and AM06, as indicated by pressure spikes (Figure 4).

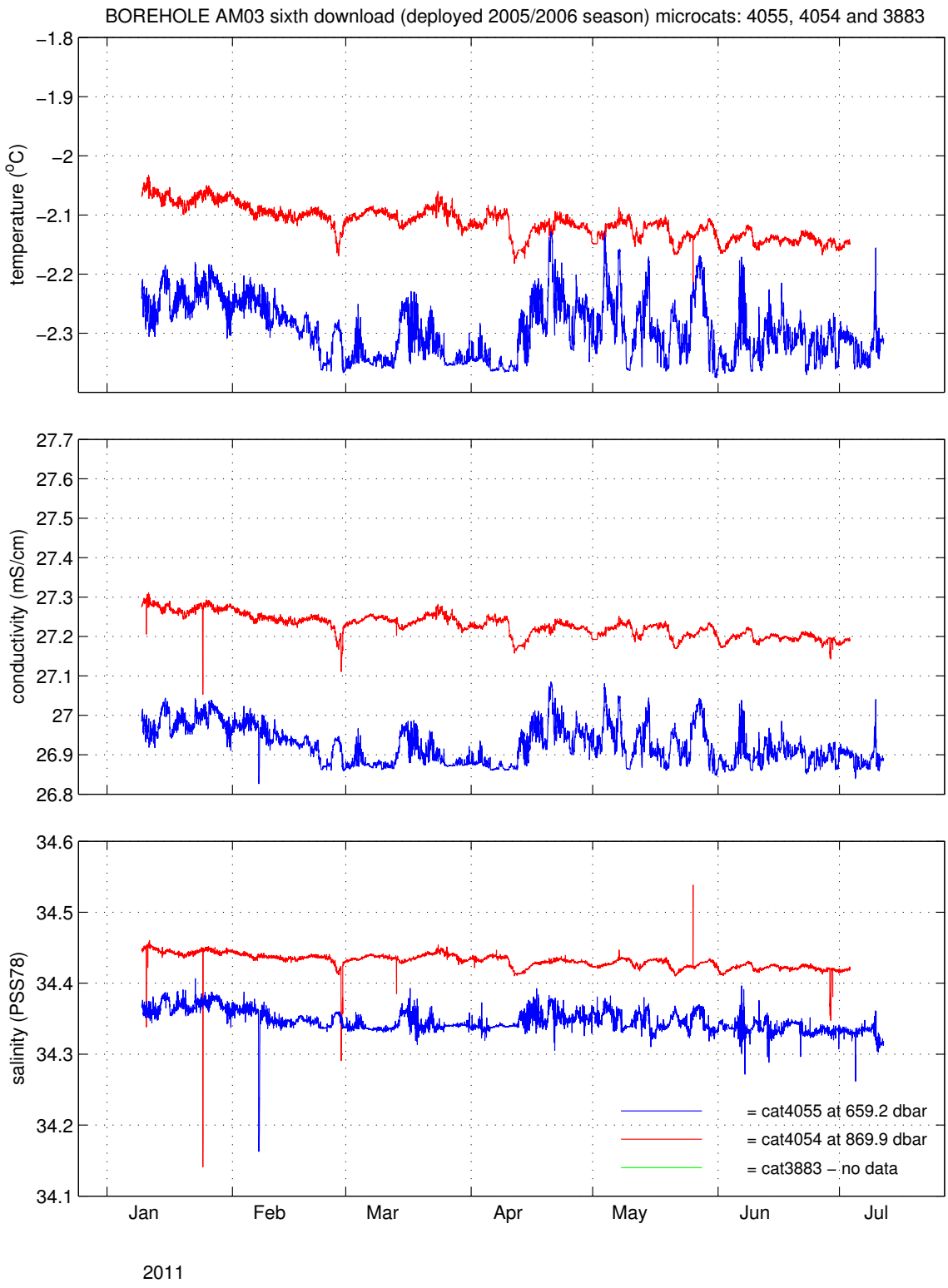
For AM03, the pressure variation throughout the time series is very similar for all 3 microcats (Figure 4), with all pressures getting shallower by ~0.3 dbar over the 6 month times series. Although the order of this drift is of the same order as the expected pressure sensor drift as quoted by the manufacturer, the strong correlation in drift between the 2 instruments suggests that the drift is real.

For AM05, there is a strong correlation between microcats 7326 and 7322 for pressure drift and pressure step events (Figure 4), indicating these features are real. For microcat 7327, all pressure data are bad, and a single value from soon after deployment has been used to recalculate all conductivity and salinity data. Use of this single pressure value obviously misses the pressure events seen in the other 2 instruments (Figure 4), and as a result the pressure for cat7327 is almost certainly in error by ~23 dbar by the end of the time series (corresponding to the cumulative drift/shifts seen in the other 2 instruments). This is equivalent to a salinity error of ~0.02 (PSS78). Note also that unexpected salinity and conductivity trends are evident in the recalculated cat7327 data, possibly real data measured by an instrument frozen in a cavity in the ice.

For the shallowest microcat (#7325) at AM06, there is a steady pressure drift similar to the previous season's data, although the magnitude of the drift (~15 dbar) is twice that of the previous season (Figure 5). There is no evidence of this drift in the other microcats, and as such it cannot be real. Note that pressure sensitivity appears unaffected. The pressure trend was removed from the data (assuming a linear trend), and initial pressure was set to the value immediately after deployment in January 2012. Conductivity/salinity values were then recalculated. Interpretation of the drifting pressure is difficult. From the results in Figure 4 (showing corrected data), there is a small variation from linear drift throughout the year, of the order 2 dbar. If the initial pressure value after deployment in January 2012 is believed, then this variation may represent an pressure error up to 2 dbar, equivalent to a salinity error of ~0.002 (PSS78).

In addition to any errors described above, from conductivity and temperature sensor calibration drift estimates as specified by the manufacturer (0.0003 (PSS78) **per month** for salinity, and 0.0005^{o}C **per year** for temperature), we might expect the following calibration drifts at the time of the latest 2011/2012 data downloads:

AM03 (in water for 6 years) total S drift = 0.022 (PSS78) total T drift = 0.003^{o}C
AM05/AM06 (in water for 2 years) total S drift = 0.007 (PSS78) total T drift = 0.001^{o}C



<u>Figure 1:</u> Borehole AM03 microcat data downloaded in 2011/2012 season. Note that there are no data for cat3883 (batteries flat).

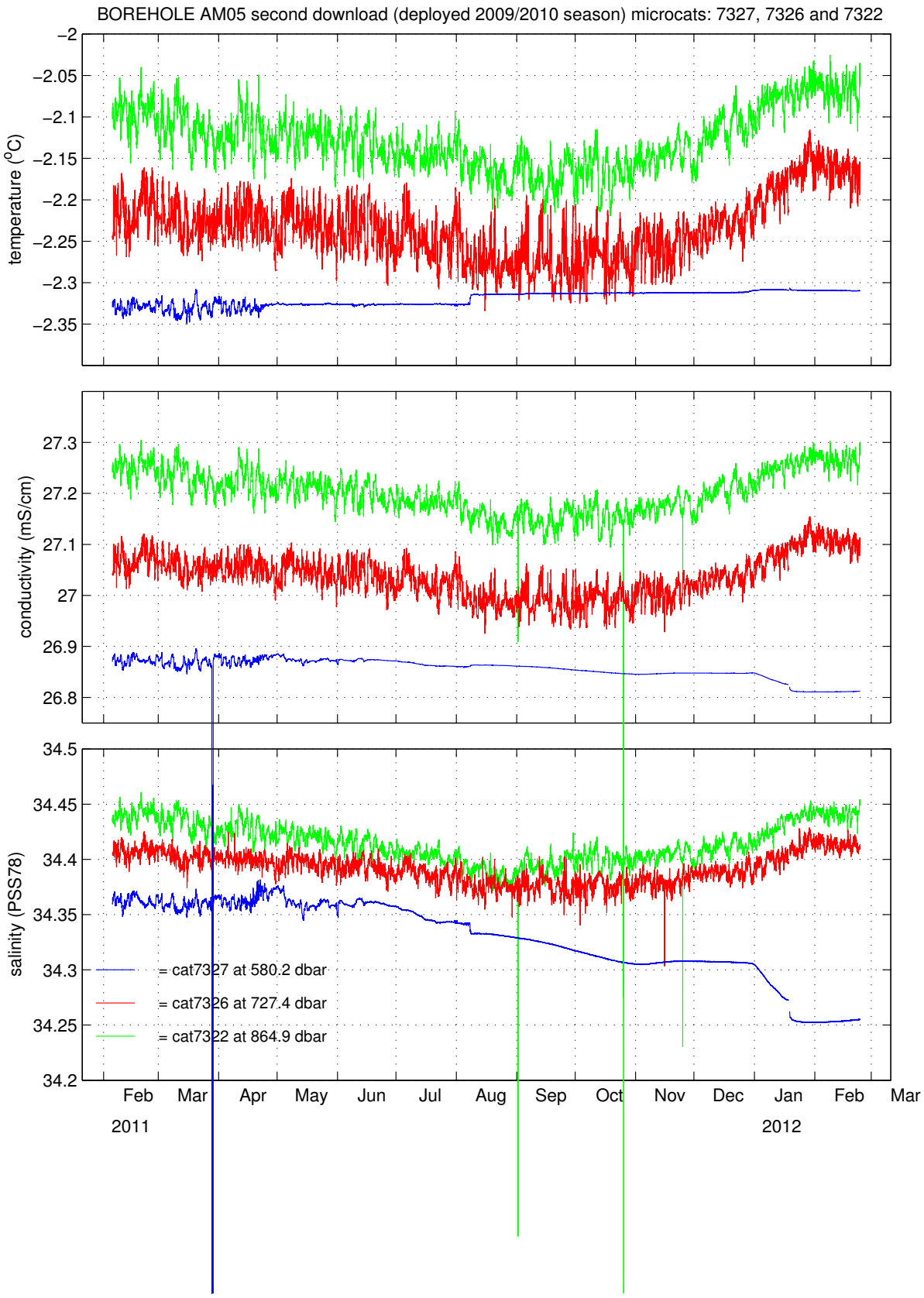


Figure 2: Borehole AM05 microcat data downloaded in 2011/2012 season.

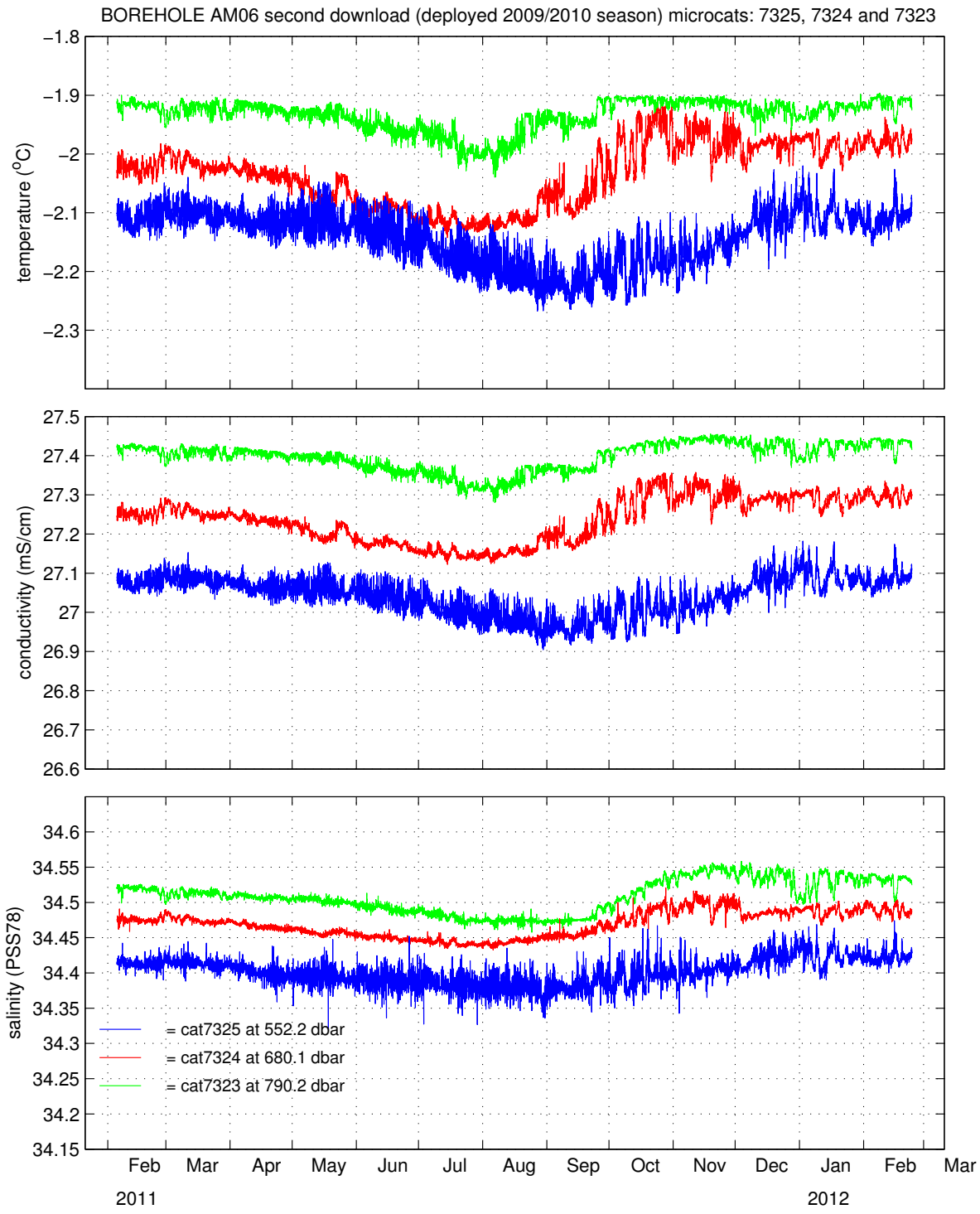
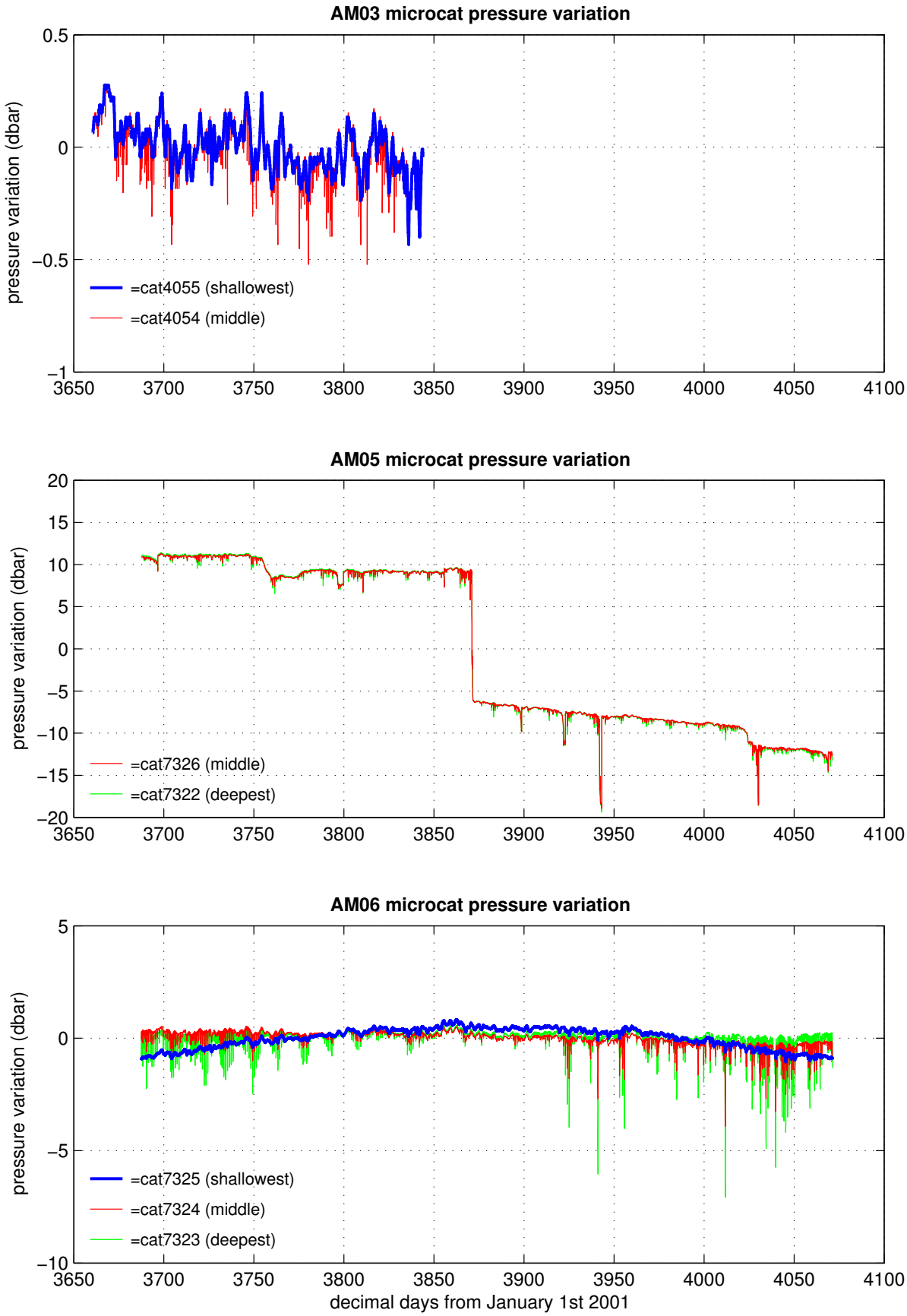
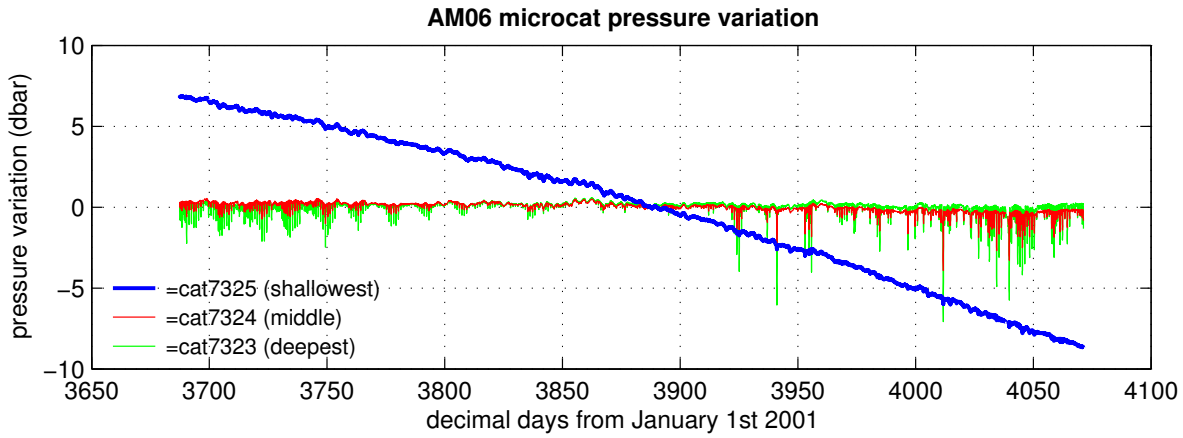


Figure 3: Borehole AM06 microcat data downloaded in 2011/2012 season.



<u>Figure 4:</u> Pressure sensor data from boreholes AM03, AM05 and AM06 microcats downloaded in 2011/2012 season. Plotted pressures are variations about the mean value. Note: pressures for AM06 shallowest microcat are corrected.



<u>Figure 5:</u> Pressure sensor data from borehole AM06 microcats, showing uncorrected pressure for shallowest microcat. Plotted pressures are variations about the mean value.

ADCP DATA

ADCP battery power was low, and the instrument was not left logging the previous season. During the AM06 site visit on 23/01/2012, 3 x 30 minute ensembles were obtained from the ADCP (Figure 6). Instrument set-up details are as per the previous season (see 2010/11 data report).

The RDI program WinADCP was used to export data from the raw files into matlab, with data in engineering units. A matlab routine was then run to change the format of matlab matrices and vectors, calculate decimal time (referred to the same time base as for the microcats i.e. decimal days as counted from midnight on December 31st 2000), replace null values with NaN, convert all direction/attitude/temperature parameters to degrees, convert all speed/velocity parameters to cm/s, and convert all depth and bottom track values to metres. Note that there were no data gaps. Also note that no clock check or clock set information was available. All ADCP parameters are averaged over each 30 minute averaging period, and in the final data files the time assigned to each record is at the ***start*** of the averaging period.

Next, quality control was applied to the data, as follows. Speed, direction, and u/v/w component bin values are flagged as bad (i.e. replaced with NaN) when:

* pcntgd4 < 80% (where pcntdgd4 = percent good of 4 beam solutions used in making the ensemble);
* average beam correlation < 70;
* orientation flag = 0 (no cases of this found in the data);
* bins 27, 26, 25, 24 and 23 all flagged as bad.

Additional notes:

* no good “bottom track” (i.e. measurement of the iceshelf underside by the upward looking transducers) data were obtained from the 3 ensembles.

* In the final matlab files, rows 1 to 27 in all matlab matrices and vectors correspond to vertical bins 27 to 1 (i.e. row 27 = bin 1, which is the deepest bin for an upward looking ADCP).

* Instrument serial number is unknown, and in the final matlab files all matrices and vector names are prefixed with “111”, as a default instrument serial number

* From pitch and roll data, the ADCP appears to be at least 2° off vertical.

The instrument pressure sensor data appear to be out of calibration (as per the previous season), and all bin depth values have been referred to the assumed instrument depth of 634 m (from the original mooring diagram). Note that the depth parameter error does not affect any other parameters.

The local magnetic declination correction at AM06 on 23/01/2012 (i.e. ADCP data measurement date) is -76.20^{o} (calculated using using the International Geomagnetic Reference Field IGRF11, as modelled by the International Association of Geomagnetism and Aeronomy Working Group V-MOD, and using the program geomag70 downloaded from the NOAA website www.ngdc.noaa.gov). In the previous year's data. the instrument data set-up files indicated a heading bias of -76° was applied to the data at the time of logging. For the current season's data, the set-up file shows a heading bias of 0° (i.e. no local magnetic declination applied), however there is no other information from field notes or field operators to confirm this. As a result, no local field correction has been applied to the data, ***but beware: it may indeed need to be applied.***

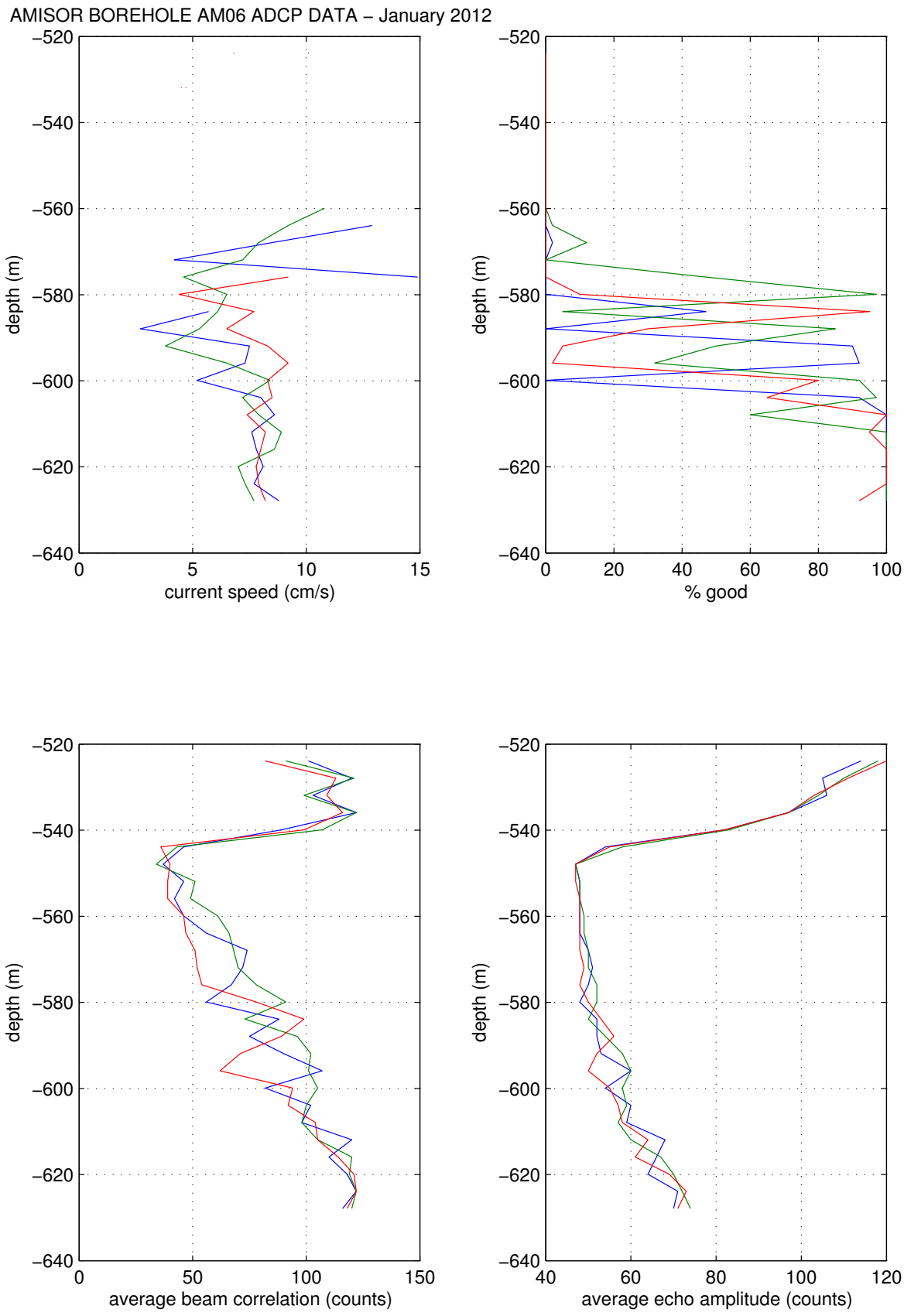


Figure 6: **ADCP data from borehole AM06 (3 ensembles only, on 23/01/2012).**

DATA FILES

New microcat data from boreholes AM03, AM05 and AM06, and ADCP data from borehole AM06, are contained in text and matlab format files, as follows:

1. Microcat data

Note: at the time of writing, the unified microcat files (i.e. all data from initial deployment time) have not been updated. New data files only contain the latest downloaded data (2011/12).

AM03

Data from the sixth data download (i.e. 09/01/2011-11/07/2011) are contained in the files:
cat4055_6.dat, cat4054_6.dat (text files), and
cat4055_6.mat, cat4054_6.mat (matlab files).
READMEcatam03_6 details the microcat file formats.

AM05

Data from the second data download (i.e. 05/02/2011-24/02/2012) are contained in the files:
cat7327_2meanp.dat, cat7326_2.dat, cat7322_2.dat (text files), and
cat7327_2meanp.mat, cat7326_2.mat, cat7322_2.mat (matlab files).
READMEcatam05_2 details the microcat file formats.
Data in these cat7327 "meanp" files use a single pressure value (from soon after initial deployment) for all salinity and conductivity calculations
The original bad pressure data, and the corresponding salinity and conductivity values, are in cat7327_2.dat and cat7327_2.mat. These data should not be used.

AM06

Data from the second data download (i.e. 05/02/2011-24/02/2012) are contained in the files:
cat7325_2.dat, cat7324_2.dat, cat7323_2.dat (text files), and
cat7325_2.mat, cat7324_2.mat, cat7323_2.mat (matlab files).
READMEcatam06_2 details the microcat file formats.
Data in these cat7325 files are corrected for the pressure drift.
Uncorrected data are in cat7325_2uncorrected.dat and cat7325_2uncorrected.mat
Data using a single pressure value (from soon after initial deployment) for all salinity and conductivity calculations are in the files cat7325_2meanp.dat and cat7325_2meanp.mat.

2. ADCP

AM06

Data from the ADCP at borehole AM05 are contained in the files:
adcp_am06_2qc.mat matlab file with the quality controlled data
adcp_am06_2noqc.mat matlab file with the complete data (including bad data) i.e. no quality control applied
READMEadcpam06_2 a README file detailing file formats

3. Station and summary info files

The files station.dat and borehole_data_summary.dat are updated to include the latest season's data.